

# **ONLINE FOOD ORDERING SYSTEM (YUM-YUM)**

**A PROJECT REPORT**

*Submitted by*

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*in partial fulfillment for the award of the degree*

*of*

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**INTEGRAL UNIVERSITY LUCKNOW**

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# INTEGRAL UNIVERSITY



## CERTIFICATE

Certified that this project report **“ONLINE FOOD ORDERING SYSTEM (YUM-YUM)”** is the bonafide work of **“MR. SOHAIM ASLAM AZMI & MR. AYAN AHMAD”** who carried out the project work under my supervision.

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## CERTIFICATE

Certified that this project report **“ONLINE FOOD ORDERING SYSTEM (YUM-YUM)”** is the bonafide work of **“MR. SOHAIM ASLAM AZMI & MR. AYAN AHMAD”** who have successfully carried out the project.

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## DECLARATION

“I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text”.

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**MR. SOHAIM ASLAM AZMI**

**MR. AYAN AHMAD**

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## **ABSTRACT**

An Online Food Ordering System is proposed here which simplifies the food ordering process. The proposed system shows a user interface and update the menu with all available options so that it eases the customer work. Customer can choose more than one item to make an order and can view order details before logging off. The order confirmation is sent to the customer. The order is placed in the queue and updated in the database and returned in real time. This system assists the staff to go through the orders in real time and process it efficiently with minimal errors.

Our proposed system is an online food ordering system that enables ease for the customers. It overcomes the disadvantages of the traditional queueing system. Our proposed system is a medium to order online food hassle free from restaurants as well as mess service. This system improves the method of taking the order from customer. The online food ordering system sets up a food menu online and customers can easily place the order as per their wish. Also, with a food menu, customers can easily track the orders. This system also provides a feedback system in which user can rate the food items. Also, the proposed system can recommend hotels, food, based on the ratings given by the user, the hotel staff will be informed for the improvements along with the quality. The payment can be made online or pay-on-delivery system. For more secured ordering separate accounts are maintained for each user by providing them an ID and a password.

## INTRODUCTION

The online food ordering system sets up a food menu online and customers can easily place the order as per they like. Also, with a food menu, online customers can easily track the orders. The management maintains customers database, and improve food delivery service. The Restaurant management systems motivates us to develop the system. There are various facilities provided so that the users of the system will get service effectively. Also, the system considers Restaurants as well as Mess facility to the customers. Again, the idea comes that mostly mess users are person who are shifted for various reason in new cities. So, they are interrelated. Increasing use of smart phones is also considered as a motivation, so that any users of this system get all service on single click. Another motivation can be considered as the system will be designed to avoid users doing fatal errors, users can change their own profile, users can track their food items through GPS, users can provide feedback and recommendations and can provide reviews.

In recent years, the advent of technology has transformed many aspects of our lives, including how we order and experience food. One of the most significant advancements in the food industry is the emergence of online food ordering systems. These platforms have revolutionized the way people order meals, offering unparalleled convenience, accessibility, and efficiency. By leveraging the power of the internet and mobile technology, online food ordering systems have become a staple in the modern dining landscape, reshaping the traditional modes of food procurement and consumption.

In the following sections, we will provide a comprehensive overview of the theoretical foundations, technical components, experimental results, and practical implications of customer services, employee services, ordering and delivery services, aiming to offer valuable insights and inspirations for developers and researchers.

This introduction sets the stage for the seminar report, outlining the ability to revolutionize the way people order and experience customer satisfaction, the objectives of the project, and what readers can expect to learn from the report.



## OBJECTIVE

The main objective of this project is to develop an application which gives provision to the restaurant owners to flourish their business by uploading menus at no cost and will invariably lead to high customer retention and acquisition rates.

- 1. Reliability:** Build a reliable and robust system that can handle high traffic volumes, process orders efficiently, and maintain uptime during peak hours. Implement measures to prevent system failures, errors, and delays that could impact the user experience.
- 2. Security:** Prioritize the security of user data, payment information, and transactions. Implement encryption, authentication, and secure payment gateways to protect sensitive information and prevent unauthorized access or fraud.
- 3. Customer Satisfaction:** Focus on delivering exceptional customer service and satisfaction. Provide responsive customer support channels, handle inquiries and complaints promptly, and address issues related to orders, deliveries, or payments effectively.
- 4. Very Less Paper Work:** Online Food Ordering System requires less paper works. In this project all record is fetched directly into the computer and reports can be generated through just a click to save time.
- 5. Personalization:** Offer personalized recommendations and promotions based on user preferences, order history, and location. Allow users to save favorite orders, customize preferences, and receive notifications about special offers and discounts.
- 6. Accuracy and Transparency:** Ensure accurate and up-to-date information about menu items, prices, availability, and delivery times. Provide transparent pricing, including taxes, fees, and delivery charges, to avoid surprises at checkout.

## **MODULES**

- Administrator Module
- Customer Module

### **Administrator Module:**

Admin can view all the information about the user to edit all the details about the customer.

- Create food category
- Manage food categories
- Add food item
- Manage food item
- Manage user order

### **Customer Module:**

These functionalities are provided:

- View product's list
- Register
- Place orders

## **Admin Module:**

- 1. Restaurant Management:** Admins can oversee restaurant profiles by adding, editing, or removing details like name, location, contact information, operating hours, and menu items.
- 2. Menu Management:** Admins have control over restaurant menus, allowing them to add new dishes, update prices, descriptions, and availability, and categorize items for ease of navigation.
- 3. Order Management:** Admins can efficiently handle orders, including tracking order status, managing order updates, and processing refunds or cancellations when required.
- 4. User Management:** Admins can manage user accounts effectively, including tasks such as registering new users, updating profiles, resetting passwords, and adjusting permissions for staff members.
- 5. Analytics and Reporting:** Admins gain insights into various metrics such as order volumes, sales revenue, popular menu items, customer feedback, and delivery times through detailed reports, aiding in informed decision-making.
- 6. Search Order:** In this section, admin can search for particular order with the help of order number.
- 7. Reports:** In this section, the admin can view order details, order counts and sales report according to dates. Admin can also update his profile, change passwords, recover passwords and accounts.

## **User Module:**

- 1. Account Registration and Profile Management:** Users can create accounts and manage personal details, including contact information and delivery addresses, to customize their experience and preferences.
- 2. Browsing and Ordering:** Users can explore restaurant menus, view dish details, and place orders for delivery or pickup, with options for customization and adding special instructions.
- 3. Search and Filter:** Users can search for restaurants and menu items based on criteria like cuisine type, location, and price range, using filter options to refine their choices for easier decision-making.
- 4. Cart Management:** Users can add items to their shopping carts, adjust quantities, and review orders before checkout, ensuring accuracy and completeness of their selections.
- 5. Checkout and Payment:** Users are guided through secure checkout processes, choosing payment options and providing delivery details, with transactions handled securely to complete orders.
- 6. Order Tracking and History:** Users can monitor the status of their orders in real-time, receiving updates on preparation, delivery, and estimated arrival times, and access a history of past orders for reference.
- 7. Feedback and Reviews:** Users can provide feedback and reviews for restaurants and menu items, sharing their experiences to help others make informed decisions and provide valuable insights for restaurant improvement.

## **Brief Information about homepage:**

In this page, guest user (which is not registered) can view restaurants food menus, search food according to food names and guest users can also sign up in restaurants and registered users can sign in. Users can also track their order without login.

# **LITERATURE SURVEY**

## **Aim**

The aim of the literature survey is to explore existing research and systems related to online food ordering, identify their strengths and weaknesses, and propose enhancements or novel approaches to improve the efficiency, user experience, and functionality of online food ordering systems.

## **Existing System**

In the existing system, online food ordering platforms typically offer basic functionalities such as browsing menus, placing orders, and making payments. These systems often lack advanced features such as personalized recommendations, real-time order tracking, and seamless integration with external services. User feedback may highlight issues with usability, order accuracy, or delivery times. Furthermore, security concerns related to data privacy and payment processing may be raised.

## **Proposed System**

The proposed system aims to address the limitations of existing systems by introducing innovative features and enhancements. This includes personalized recommendations based on user preferences and order history, real-time order tracking using GPS technology, and integration with third-party services such as food delivery aggregators and payment gateways to improve convenience and reliability. Additionally, measures to enhance security and privacy, such as data encryption and secure payment processing, will be implemented to ensure user trust and confidence in the system. Overall, the proposed system aims to provide a more seamless, user-friendly, and secure online food ordering experience for customers.

# DESIGN

## INTRODUCTION:

The design phase of the online food ordering system involves conceptualizing the architecture, user interface, and functionality of the system. It lays the foundation for the development process by outlining the system's components, interactions, and workflows.

## Uml Diagrams:

UML (Unified Modeling Language) diagrams are used to visually represent the system's structure and behavior. This includes diagrams such as:

- **Use Case Diagrams:** Illustrate the system's functionality from the perspective of different actors (e.g., users, administrators).
- **Class Diagrams:** Depict the static structure of the system, including classes, attributes, and relationships.
- **Sequence Diagrams:** Show the interactions between system components or actors over time, depicting the flow of messages or actions.
- **Activity Diagrams:** Represent the workflow or process flow within the system, detailing the sequence of activities and decision points.

## Use case:

Use cases describe the interactions between users (actors) and the system to achieve specific goals or tasks. In the context of an online food ordering system, typical use cases include:

- **Registering/Login:** Users create accounts or log in to access the system.
- **Browsing Menu:** Users explore restaurant menus and view dish details.
- **Placing Order:** Users select items, customize orders, and proceed to checkout.

- **Managing Profile:** Users update personal information, preferences, or payment methods.
- **Admin Management:** Administrators manage restaurants, menus, orders, and user accounts.
- **Order Tracking:** Users track the status of their orders in real-time.

There are various kinds of methods in software design:

They are as follows:

- Use case diagram
- Sequence diagram
- Collaboration diagram
- Activity diagram
- State chat diagram

## USECASE DIAGRAMS:

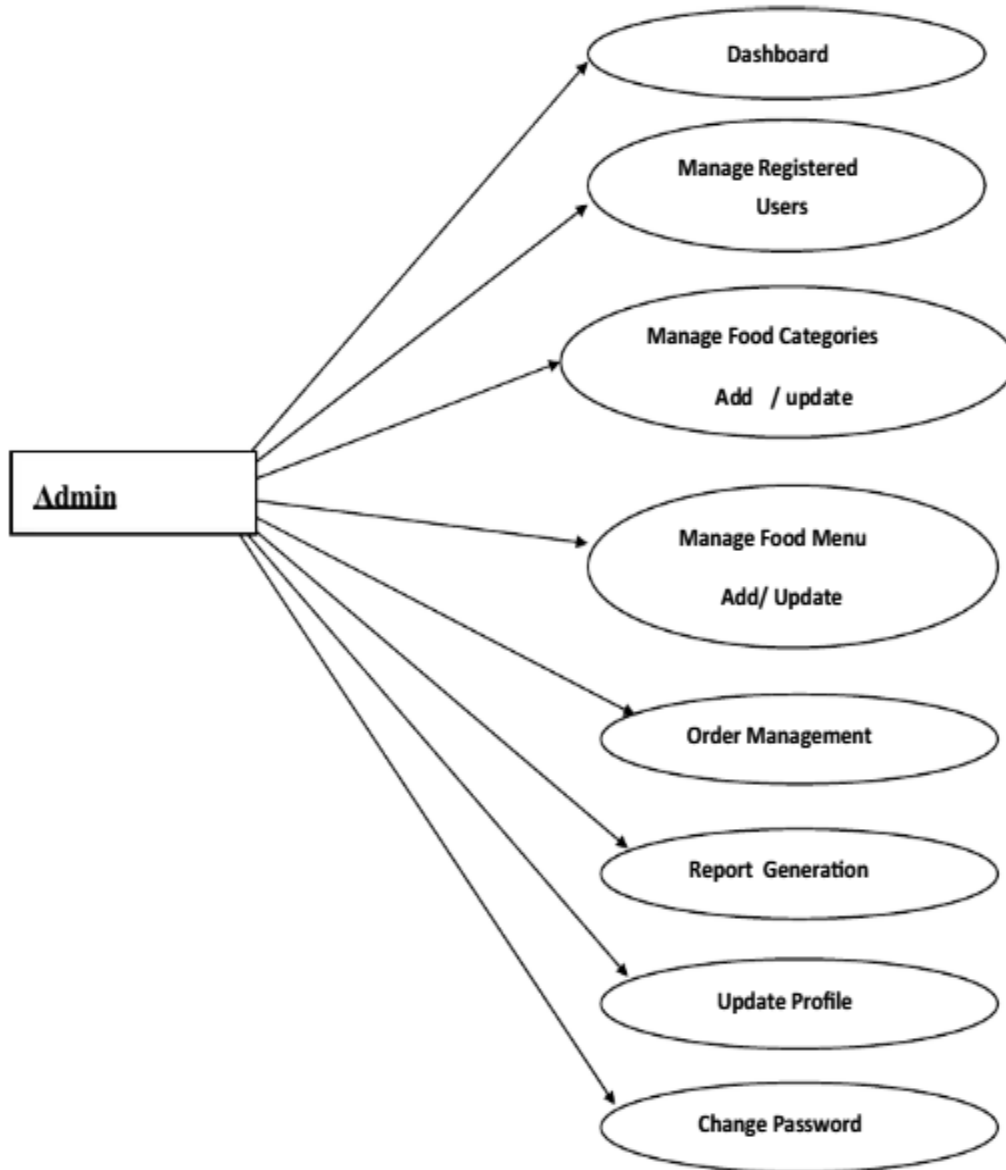
Use case diagrams model behavior within a system and helps the developers understand of what the user requires. The stickman represents what's called an actor.

Use case diagram can be useful for getting an overall view of the system and clarifying who can do and more importantly what they can't do.

Use case diagrams consists of use cases and actors and shows the interaction between the use case and actors.

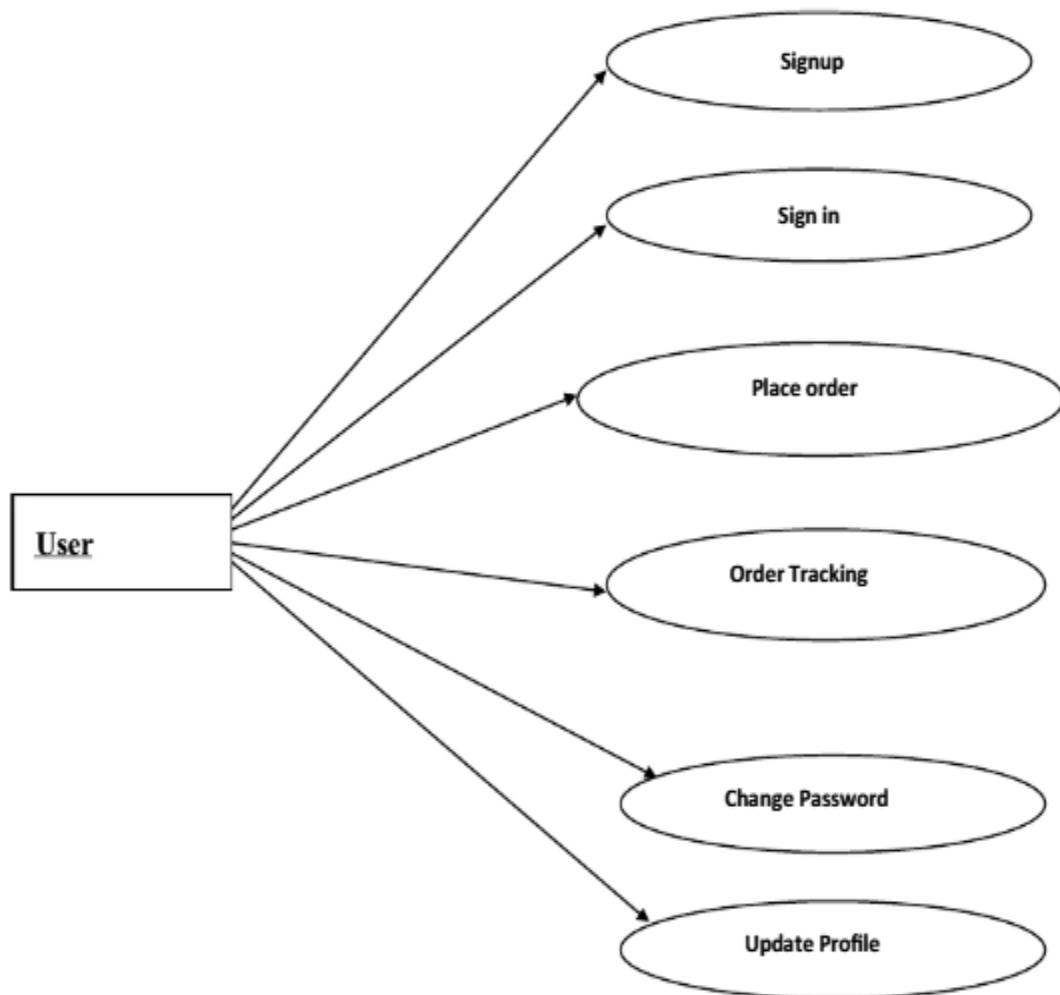
- The purpose is to show the interactions between the use case and actor.
- To represent the system requirements from user's perspective.
- An actor could be the end-user of the system or an external system.

## Admin Use Case Diagram:





## User Use Case Diagram:



## **HARDWARE AND SOFTWARE REQUIREMENT**

### **Software Requirements:**

Windows 10 or higher

### **Languages Used:**

**Front-End:** HTML, JavaScript, CSS

**Back-End:** PHP

**Database Management:** MySQL

**Operating System:** The script should be compatible with various operating systems like Windows, macOS, and Linux, as long as there is stable browser with internet connection.

### **Softwares Used:**

**Web Servers:** XAMPP, WAMPP

**IDEs:** Visual Studio Code, Notepad ++

**Browser:** Chrome, Firefox

### **Hardware Requirements:**

**Processor:** Any Intel Core i3 or AMD Ryzen 3

**Processor Speed:** 2.60 GHz

**Graphics Card:** AMD Radeon

**Ram:** 1 GB (minimum)

**Hard Disk:** 10 GB (minimum) HDD/SSD

**Platform:** Windows 7/8/10/11 64-Bit

## FRONT-END

### HTML:

HTML, or Hypertext Markup Language, is the standard markup language used to create and design web pages. It provides the basic structure and content of a webpage by using a system of tags and elements. HTML allows developers to define the layout, format text, embed images, create links, and add multimedia content such as audio and video. It serves as the foundation of web development, providing the structure upon which websites are built and enabling users to access and interact with information on the internet.

1. **Creating Links:** HTML allows developers to create hyperlinks that connect different pages within the website. For example, links can be used to navigate between restaurant listings, menu pages, and the checkout process.
2. **Embedding Images:** HTML provides tags for embedding images into web pages. In an online food ordering system, images of dishes, restaurant logos, and promotional banners can be displayed using HTML img tags to enhance visual appeal and attract users.
3. **Displaying Text Content:** HTML is used to display text content such as restaurant descriptions, menu items, prices, and order details. Text can be formatted using HTML tags like headings, paragraphs, lists, and emphasis tags to improve readability and presentation.
4. **Implementing Forms:** HTML forms are essential for collecting user input, such as delivery addresses, customization options, and payment details. Forms allow users to interact with the website by entering information and submitting orders securely.
5. **Structuring Content:** HTML is used to organize the content of the website into logical sections such as header, navigation, main content, and footer. This helps users navigate the site efficiently and find the information they need.

## JAVASCRIPT:

JavaScript is a versatile programming language commonly used in web development to add interactivity and dynamic functionality to web pages. It runs on the client-side, meaning it executes code within the user's web browser, enabling actions like form validation, animations, and asynchronous communication with servers.

- 1. Form Validation:** JavaScript ensures data accuracy by validating user input in forms, checking for errors such as missing fields or incorrect information, and providing instant feedback to users, preventing submission of incomplete or erroneous orders.
- 2. Dynamic Content Updates:** JavaScript updates content dynamically without page reloads, such as updating the shopping cart or displaying real-time order status updates, providing a seamless and responsive user experience.
- 3. Asynchronous Requests (AJAX):** JavaScript facilitates background communication with the server, allowing for asynchronous data retrieval without disrupting user interaction. This enables features like live search, auto-complete suggestions, and dynamic loading of restaurant listings and menu items.
- 4. User Authentication and Authorization:** JavaScript implements user authentication and authorization mechanisms, enabling registered users to log in, manage accounts, and access personalized features like order history and saved preferences, enhancing security and personalization.
- 5. Integration with Third-Party APIs:** JavaScript integrates with third-party APIs for additional functionalities, such as payment gateways for secure transactions, geolocation APIs for address validation and delivery tracking, and social media APIs for sharing order details, expanding the system's capabilities and providing enhanced user experiences.
- 6. User Interaction:** JavaScript creates interactive elements like dropdown menus, pop-up dialogs, and forms, enhancing user engagement and improving the ordering process's interactivity.

## CSS:

CSS or Cascading Style Sheets is essential for creating a visually appealing, user-friendly, and accessible online food ordering system that effectively conveys the brand's identity and provides a seamless experience for users.

- 1. Layout and Structure:** CSS defines the layout and arrangement of elements, ensuring consistency and visual appeal across the website's pages.
- 2. Typography:** CSS styles text content, improving readability and user experience by customizing fonts, sizes, and alignments.
- 3. Color Scheme and Visual Elements:** CSS creates a cohesive design by defining color schemes, backgrounds, borders, and other visual elements, reinforcing branding and identity.
- 4. Responsive Design:** CSS enables the website to adapt to different devices and screen sizes, providing a seamless experience for users on desktops, tablets, and smartphones.
- 5. Animations and Transitions:** CSS adds interactivity and visual interest through animations and transitions, enhancing user engagement and feedback during the ordering process.
- 6. Accessibility:** CSS contributes to accessibility by styling elements for readability and usability, ensuring the website is inclusive and usable for all users, including those with disabilities.

## BACK-END

### **PHP:**

PHP, which stands for Hypertext Preprocessor, is a widely-used server-side scripting language primarily designed for web development. It's often embedded into HTML to add dynamic features to web pages or interact with databases. PHP code is executed on the server, generating HTML content that is then sent to the client's web browser. It's known for its simplicity, versatility, and compatibility with various databases and web servers. PHP is commonly used for tasks like handling form data, generating dynamic page content, managing user sessions, and interacting with databases, making it a fundamental component of many websites and web applications.

- 1. Processing Form Data:** PHP handles user submissions like orders and payment info, validating and securely storing the data.
- 2. Generating Dynamic Content:** PHP fetches and displays real-time content like menus and promotions from databases, ensuring users see updated information.
- 3. Managing User Sessions:** PHP manages user logins and personalized features, maintaining session data for seamless browsing experiences.
- 4. Interacting with Databases:** PHP communicates with databases to store and retrieve essential data, ensuring consistency and integrity.
- 5. Handling Business Logic:** PHP executes critical business operations such as calculating totals, validating inputs, and managing discounts.
- 6. Integrating with External Services:** PHP integrates with external services like payment gateways and geolocation APIs to enhance system functionality, ensuring secure transactions and efficient order management.

## DATABASE MANAGEMENT

### MySQL:

MySQL is a popular open-source relational database management system (RDBMS) that is widely used in web development. It provides a powerful and scalable platform for storing, organizing, and retrieving data in a structured manner. MySQL uses SQL (Structured Query Language) as its primary interface for managing databases, allowing users to perform operations such as querying, updating, and deleting data. It supports features like transactions, indexing, and replication, making it suitable for a wide range of applications, from small-scale websites to large enterprise systems.

- 1. Storing User Data:** MySQL is used to store user information such as login credentials, profile details, and order history. This data allows users to log in, manage their accounts, and access personalized features.
- 2. Recording Orders:** MySQL records details of user orders, including selected items, quantities, special instructions, and order statuses. This data facilitates order processing, tracking, and fulfillment, ensuring efficient handling of customer requests.
- 3. Managing Restaurant Information:** MySQL stores information about partner restaurants, including names, addresses, contact details, operating hours, and delivery areas. This data helps users find nearby restaurants, view their details, and place orders for delivery or pickup.
- 4. Tracking Delivery Information:** MySQL stores delivery-related information such as customer addresses, delivery statuses, delivery personnel assignments, and delivery times. This data enables real-time tracking of delivery orders and ensures timely and accurate delivery to customers.
- 5. Handling Payment Transactions:** MySQL records transaction details for online payments, including payment methods, amounts, transaction IDs, and payment statuses. This data facilitates secure and reliable processing of payment transactions, ensuring smooth and seamless checkout experiences for users.

## PROBLEM IDENTIFICATION AND FEASIBILITY STUDY

### Problem Identification:

1. **Security Concerns:** Users may be hesitant to provide personal information or payment details due to concerns about data privacy, security breaches, or fraudulent activities.
2. **Delivery Issues:** Late deliveries, missing items, or poor packaging quality can lead to negative reviews and damage the reputation of the system and its partner restaurants.
3. **Limited Restaurant Options:** A limited selection of restaurants or cuisines may reduce user satisfaction and limit the system's appeal to a wider audience.
4. **Lack of Integration:** Poor integration with third-party services such as delivery partners, payment gateways, or inventory management systems can lead to inefficiencies and errors in order processing.
5. **Scalability Challenges:** Difficulty in handling increased traffic, orders, or user accounts during peak times can lead to system slowdowns or failures, impacting user experience and revenue generation.
6. **Limited Features:** The system may lack essential features such as order tracking, user reviews, personalized recommendations, or integration with popular payment methods, limiting its functionality and usability.
7. **Inaccurate Information:** Incorrect or outdated menu items, prices, or restaurant details can lead to misunderstandings, order errors, and customer dissatisfaction.
8. **Poor Communication:** Lack of communication between users and customer support, delayed responses to inquiries, or insufficient updates on order status can result in dissatisfaction and loss of trust.



## Feasibility Study:

### 1. Technical Feasibility:

- **Available Technology:** Evaluate the existing technology stack and infrastructure to determine if it can support the development and operation of the system. Consider factors such as web development frameworks, databases, hosting platforms, and scalability requirements.
- **Development Tools:** Assess the availability and suitability of development tools and resources needed to build the system, including programming languages, integrated development environments (IDEs), version control systems, and third-party APIs.
- **Expertise:** Evaluate the technical expertise and skills required to implement and maintain the system. Determine if in-house resources possess the necessary knowledge or if external expertise is needed to fill any gaps.

### 2. Operational Feasibility:

- **Staff Training:** Assess the readiness of staff members to adopt and use the online food ordering system effectively. Determine if training programs or support materials will be required to familiarize users with the system's features and functionality.
- **Workflow Integration:** Evaluate how the system will integrate with existing operational workflows and processes within restaurants, delivery services, and customer support teams. Identify any potential bottlenecks or challenges that may arise during implementation.
- **Disruption Mitigation:** Develop strategies to minimize disruptions to existing processes and ensure a smooth transition to the new system. Consider phased implementation approaches, pilot testing, and contingency plans to address unforeseen issues.

### 3. Financial Feasibility:

- **Development Costs:** Estimate the initial investment required to develop the online food ordering system, including expenses for software development, hardware infrastructure, and licensing fees.
- **Ongoing Maintenance Expenses:** Calculate the recurring costs associated with maintaining and supporting the system, including server hosting, software updates, bug fixes, and technical support.
- **Revenue Streams:** Identify potential sources of revenue, such as transaction fees, subscription models, advertising revenue, or partnerships with restaurants and delivery services. Evaluate the projected return on investment (ROI) and profitability of the system over time.

### 4. Market Feasibility:

- **Demand Analysis:** Conduct market research to assess the demand for online food ordering services within the target demographic. Analyze consumer preferences, behavior patterns, and willingness to adopt digital platforms for ordering food.
- **Competitor Analysis:** Evaluate the competitive landscape by analyzing existing online food ordering platforms, their features, pricing models, and market share. Identify opportunities to differentiate the system and attract users.
- **Trend Analysis:** Monitor emerging trends in the food delivery industry, such as changes in consumer preferences, technology adoption, and regulatory developments. Anticipate future market dynamics and adapt the system accordingly.

## **5. Legal and Regulatory Feasibility:**

- **Data Privacy Laws:** Ensure compliance with data privacy regulations, such as the General Data Protection Regulation (GDPR) in Europe or the California Consumer Privacy Act (CCPA) in the United States. Implement measures to protect user data and obtain necessary consent for data processing.
- **Food Safety Regulations:** Adhere to food safety standards and regulations imposed by local health authorities or government agencies. Ensure that restaurants and delivery partners comply with food handling, storage, and sanitation requirements.
- **Intellectual Property Rights:** Respect intellectual property rights related to trademarks, copyrights, and patents when developing and branding the online food ordering system. Obtain necessary licenses or permissions for using third-party content or software components.

## **OBJECTIVES**

An online food ordering system is a digital platform that enables users to browse menus, place orders, and arrange delivery or pickup of food items from a variety of restaurants and cuisines. Its objectives encompass enhancing convenience and accessibility by providing a streamlined ordering process accessible anytime, anywhere, promoting efficiency and time savings through features like saved preferences and quick reordering, expanding culinary choices with a diverse selection of restaurants and menus, prioritizing user experience with intuitive interfaces and transparent information access, ensuring order accuracy and reliability through robust management systems, and driving business growth and sustainability for restaurants by increasing their reach and optimizing operations. Overall, the system aims to deliver a seamless, efficient, and enjoyable dining experience for users while supporting the growth and success of restaurant partners.

### **1. Convenience and Accessibility:**

- Provide users with a convenient and accessible platform to browse menus, place orders, and track deliveries from a wide range of restaurants and cuisines.
- Enable users to access the system anytime, anywhere, using various devices such as smartphones, tablets, and computers, to cater to diverse lifestyles and preferences.

### **2. Efficiency and Time Savings:**

- Streamline the food ordering process, reducing wait times and eliminating the need for phone calls or physical visits to restaurants.
- Offer features such as saved preferences, order history, and one-click reordering to expedite the ordering process and enhance user efficiency.

### **3. Expanded Choice and Variety:**

- Expand users' culinary options by providing access to a diverse selection of restaurants, menus, and cuisines, including local favorites, international cuisines, and specialty dishes.
- Facilitate exploration and discovery of new dining experiences through personalized recommendations, featured promotions, and user reviews.

### **4. Enhanced User Experience:**

- Deliver a seamless and intuitive user experience across all touchpoints, including website interfaces, mobile applications, and customer support channels.
- Prioritize user-centric design principles such as simplicity, clarity, and accessibility to ensure a positive and engaging experience for all users.

### **5. Transparency and Information Accessibility:**

- Promote transparency and information accessibility by providing users with comprehensive details about restaurants, menus, pricing, delivery options, and order status.
- Enable users to make informed decisions based on factors such as food preferences, dietary restrictions, restaurant ratings, and delivery times.

### **6. Order Accuracy and Reliability:**

- Ensure order accuracy and reliability through robust order management systems, real-time inventory tracking, and quality control measures.
- Minimize errors, delays, and discrepancies in order processing, payment transactions, and delivery fulfillment to build trust and loyalty among users.

## **7. Business Growth and Sustainability:**

- Drive business growth and sustainability for restaurants and food establishments by expanding their reach, increasing order volumes, and optimizing operational efficiency.
- Offer marketing and promotional opportunities for restaurants to showcase their offerings, attract new customers, and cultivate customer loyalty through rewards programs and incentives.

# SYSTEM TESTING

## 1. Project Implementation & Testing:

- **Project Implementation:** This phase involves deploying the online food ordering system into a live environment after development is complete. Implementation includes setting up servers, databases, and other infrastructure components, configuring the system, and ensuring it is ready for use.
- **Testing:** System testing is a crucial step to verify that the online food ordering system functions as intended and meets specified requirements. Testing involves various activities such as functional testing, usability testing, performance testing, security testing, and compatibility testing to identify and resolve any issues or defects before launch.

## 2. Implementation Issues & Challenges:

- **Technical Challenges:** Implementation may face technical challenges such as integration issues with third-party services (e.g., payment gateways, delivery partners), compatibility issues with different devices and browsers, or scalability concerns as user traffic increases.
- **Operational Challenges:** Challenges related to operational aspects may include training staff members to use the system effectively, adapting existing workflows to accommodate the new system, and ensuring smooth coordination between different teams involved in order processing and delivery.
- **Regulatory Compliance:** Compliance with legal and regulatory requirements, such as data privacy laws and food safety regulations, may pose challenges during implementation. Ensuring that the system meets these compliance standards is essential to avoid legal issues and maintain trust with users.

### 3. User without background:

- **User-Friendly Design:** The online food ordering system should be designed with simplicity and intuitiveness in mind to accommodate users without a technical background. Clear instructions, intuitive navigation, and user-friendly interfaces can help users easily understand how to use the system.
- **Guided Onboarding:** Implementing guided onboarding processes, tutorials, tooltips, or help sections can assist users in learning how to use the system effectively, even if they lack prior experience with similar platforms.
- **Accessible Support Channels:** Providing accessible support channels, such as live chat, email support, or FAQ sections, can help users troubleshoot issues or get assistance when needed. Responsive customer support can alleviate concerns and ensure a positive user experience for all users, regardless of their background.

### 4. The screen size of different device:

This would be one of the issues that will encounter while implementing the system. Because user is able to download the mobile application from the official website and use it as a client device to place order. Therefore, if the user device screen sizes are too small or too big.

### 5. Server performance:

During real time system implementation, there would be a huge number of client that access to the server at the same time. Therefore, it may slow down the connection and performances of the system and even causes the server to come crashing down if the issue goes beyond the level of acceptance.



# REQUIREMENT ANALYSIS

## 1. Functional Requirements:

- **User Registration and Authentication:** Users should be able to register for an account on the platform and authenticate themselves securely to access personalized features.
- **Menu Browsing:** Users should be able to browse through menus of different restaurants, view dish details, prices, and images, and filter or search for specific items.
- **Order Placement:** Users should be able to select items from the menu, add them to their cart, specify quantity and customization options, and place orders for delivery or pickup.
- **Payment Processing:** The system should support various payment methods such as credit/debit cards, digital wallets, or cash on delivery, and ensure secure payment processing.
- **Order Tracking:** Users should be able to track the status of their orders in real-time, from preparation to delivery, and receive notifications or updates on order progress.
- **Feedback and Reviews:** Users should be able to provide feedback and reviews for restaurants and menu items, share their experiences, and rate the quality of service.
- **Admin Management:** Administrators should have access to tools for managing restaurant listings, menus, orders, user accounts, and other system settings.
- **Reporting and Analytics:** The system should generate reports and analytics on key metrics such as order volumes, sales revenue, customer feedback, and delivery performance to help stakeholders make informed decisions.

## 2. Non-Functional Requirements:

- **Performance:** The system should be responsive and scalable to handle concurrent user traffic, ensuring fast loading times and minimal downtime during peak hours.
- **Usability:** The user interface should be intuitive, easy to navigate, and accessible across different devices and screen sizes, catering to users with varying levels of technical proficiency.
- **Security:** The system should implement robust security measures to protect user data, payment information, and system integrity, including encryption, authentication, and authorization mechanisms.
- **Reliability:** The system should be reliable and resilient, with built-in redundancy and failover mechanisms to minimize service disruptions and data loss.
- **Scalability:** The system should be designed to scale horizontally or vertically to accommodate growth in user base and transaction volume without sacrificing performance or reliability.
- **Availability:** The system should be available and accessible to users 24/7, with minimal planned downtime for maintenance or upgrades, and rapid recovery from unplanned outages.
- **Compliance:** The system should comply with relevant regulations and standards, such as data privacy laws, food safety regulations, and industry best practices, to ensure legal and ethical operation.

## **REVIEW OF PREVIOUS WORK**

Previous work in the field of real-time color detection has laid the groundwork for the development of systems like the one described in this project.

### **1. Existing Techniques:**

Explore the range of existing techniques employed in online food ordering systems, including web-based platforms, mobile applications, cloud-based solutions, and integration with emerging technologies like AI, IoT, and blockchain.

### **2. Hardware and Software Platforms:**

Examine the hardware and software platforms commonly used in online food ordering systems, such as servers, databases, web frameworks (e.g., Django, Ruby on Rails), mobile operating systems (e.g., iOS, Android), and development tools (e.g., IDEs, version control systems).

### **3. Applications and Use Cases:**

Analyze the various applications and use cases of online food ordering systems, including consumer-facing platforms for browsing menus, placing orders, and tracking deliveries, as well as backend systems for restaurant management, order processing, and delivery logistics.

### **4. Challenges and Limitations:**

Identify the challenges and limitations faced by online food ordering systems, such as competition from established players, customer retention, delivery logistics, food quality control, regulatory compliance, cybersecurity risks, and scalability issues during peak demand.

### **5. Performance Evaluation:**

Evaluate the performance of online food ordering systems based on key metrics such as response times, uptime, order accuracy, delivery

speed, customer satisfaction scores, and business metrics like revenue growth, market share, and customer retention rates.

## **6. Future Directions:**

Discuss potential future directions for research and development in online food ordering systems, including the adoption of emerging technologies like AI-driven personalization, chatbots for customer support, blockchain for supply chain transparency, and sustainability initiatives to reduce food waste and environmental impact.

# METHODOLOGY

## **1. Problem Understanding:**

Begin by thoroughly understanding the requirements and challenges of online food ordering, including user needs, business objectives, market trends, and regulatory considerations. Conduct stakeholder interviews, surveys, and market research to gain insights into the problem domain.

## **2. Research and Familiarization:**

Conduct extensive research to explore existing online food ordering systems, industry best practices, emerging technologies, and potential competitors. Familiarize yourself with relevant programming languages, frameworks, and tools commonly used in this domain.

## **3. Environment Setup:**

Set up the development environment by configuring servers, databases, development tools, and version control systems. Create a robust infrastructure for hosting and deploying the online food ordering system, ensuring scalability, security, and reliability.

## **4. Code Implementation:**

Develop the online food ordering system according to the identified requirements and design specifications. Use appropriate programming languages, frameworks, and architectural patterns to implement features such as user authentication, menu management, order processing, and payment integration.

## **5. Testing and Validation:**

Conduct thorough testing to validate the functionality, usability, security, and performance of the system. Perform unit testing, integration testing, and system testing to identify and fix defects, ensuring that the system meets quality standards and user expectations.

## **6. Performance Optimization:**

Optimize the performance of the online food ordering system by analyzing bottlenecks, optimizing database queries, caching frequently accessed data, and implementing techniques such as lazy loading and code minification. Monitor system performance and scalability to ensure optimal user experience under varying loads.

## **7. Documentation and Reporting:**

Document the design, architecture, implementation details, and testing procedures of the online food ordering system. Prepare comprehensive user manuals, technical documentation, and reports to facilitate system maintenance, troubleshooting, and knowledge transfer.

## **8. Future Enhancements:**

Continuously monitor user feedback, market trends, and technological advancements to identify opportunities for future enhancements and optimizations. Prioritize and plan future development initiatives to add new features, improve user experience, and stay ahead of competition in the dynamic online food ordering market.

## **DEVELOPMENT TOOLS**

### **1. Database Environment:**

- A robust database environment is crucial for storing and managing data related to users, restaurants, menus, orders, payments, and more in the online food ordering system.
- Tools like MySQL, PostgreSQL, or MongoDB provide features for creating, querying, updating, and maintaining databases. They ensure data integrity, security, and scalability, allowing the system to handle large volumes of transactions.

### **2. Web Technology:**

- Web technologies are the foundation of the online food ordering system, powering the frontend user interface and backend server-side logic.
- HTML, CSS, and JavaScript are fundamental for building responsive and interactive user interfaces. Frameworks and libraries like React, Angular, or Vue.js streamline frontend development, while server-side technologies like Node.js, Django, or Ruby on Rails facilitate backend development.

### **3. System Platform:**

- Choosing the right system platform is essential for deploying and running the online food ordering system reliably and securely.
- Cloud platforms such as AWS, Azure, or Google Cloud offer scalable infrastructure, managed services, and security features for hosting web applications. They provide resources like virtual servers, databases, storage, and networking, enabling seamless deployment and management of the system.

#### **4. Project Management Tool:**

- Project management tools help organize tasks, track progress, collaborate with team members, and ensure timely delivery of project milestones.
- Tools like Jira, Trello, or Asana facilitate task management, issue tracking, sprint planning, and communication among team members. They provide features such as Kanban boards, Gantt charts, and workflow automation to streamline project workflows and enhance productivity.

#### **5. Visual Paradigm Community Editor:**

- Visual modeling tools are valuable for designing system architectures, data models, and software diagrams to visualize and communicate project requirements and designs.
- Visual Paradigm Community Edition offers a range of diagramming tools for creating UML diagrams, ER diagrams, flowcharts, and more. It helps developers and stakeholders understand system components, relationships, and behaviors, aiding in design decisions and implementation planning.

#### **6. Test Plan:**

- Testing is critical for ensuring the quality, reliability, and functionality of the online food ordering system before deployment. After the system has been developed, it will move to the system testing phase. In system testing phase, the developed system is required to install on appropriate devices for testing purpose. After the system install has been completed, the system testing task will be performed by different roles of user such as manager role and staff role.
  - i. **Unit Testing:** Involves testing individual components or modules of the system in isolation to verify their correctness and functionality. Tools like Jest, Mocha, or PHPUnit automate



unit tests, allowing developers to catch and fix bugs early in the development cycle.

- ii. **Functional Testing:** Focuses on validating the system's functionality against specified requirements and use cases. Tools like Selenium, Cypress, or Puppeteer automate functional tests by simulating user interactions and verifying expected outcomes, ensuring the system behaves as intended across different scenarios.

# TESTING

## 1. Unit Testing:

- **Purpose:** Unit testing verifies the correctness of individual components or modules of the system in isolation.
- **Approach:** Developers write test cases to validate the behavior of functions, classes, or methods, covering different scenarios and edge cases.
- **Tools:** Common unit testing frameworks include Jest, Mocha, PHPUnit, and Jasmine.
- **Example:** Testing functions that handle user authentication, menu retrieval, order processing, and payment validation to ensure they work as expected.

## 2. Integration Testing:

- **Purpose:** Integration testing verifies the interactions between different components or modules of the system.
- **Approach:** Developers test how components interact with each other, ensuring seamless communication and data flow.
- **Tools:** Integration testing frameworks such as Supertest, Flask-Testing, and Django's TestCase module.
- **Example:** Testing the integration between the frontend user interface and backend API endpoints to ensure proper data transmission and error handling.

## 3. System Testing:

- **Purpose:** System testing validates the entire online food ordering system as a whole.

- **Approach:** Testers simulate real-world user interactions and scenarios to verify the system's functionality, performance, and usability.
- **Tools:** Automated testing tools like Selenium, Cypress, and Puppeteer for end-to-end testing.
- **Example:** Testing the complete ordering process, including user registration, menu browsing, order placement, payment processing, and order tracking, to ensure all features work seamlessly together.

#### 4. Acceptance Testing:

- **Purpose:** Acceptance testing verifies that the system meets the specified requirements and is ready for deployment.
- **Approach:** Testers execute predefined test cases based on user stories, use cases, and acceptance criteria to validate the system's functionality from an end-user perspective.
- **Tools:** Manual testing by stakeholders, automated acceptance testing tools like Cucumber or Behave.
- **Example:** Testing the online food ordering system with real users or stakeholders to ensure it meets their expectations in terms of usability, performance, and business requirements.

#### 5. Regression Testing:

- **Purpose:** Regression testing ensures that recent code changes or updates do not introduce new bugs or regressions.
- **Approach:** Re-run previously executed test cases to ensure existing functionalities remain intact after code changes.
- **Tools:** Test automation frameworks integrated with continuous integration (CI) pipelines.

- **Example:** After implementing new features or fixing bugs, re-running all previously passed test cases to verify that no unintended side effects have occurred.

## **SYSTEM LIMITATIONS AND SYSTEM STRENGTH**

### **SYSTEM LIMITATIONS**

One significant limitation of online food ordering systems is their susceptibility to technical issues and disruptions. These systems heavily rely on technology, making them vulnerable to website crashes, slow loading times, and payment processing errors, which can impede the user experience and lead to frustration among customers. Additionally, the dependency on internet connectivity poses challenges in areas with poor network coverage or during network outages, hindering users' ability to access the platform and place orders seamlessly. These technical limitations underscore the importance of robust infrastructure, continuous monitoring, and proactive measures to ensure the reliability, stability, and performance of online food ordering systems.

### **SYSTEM STRENGTHS**

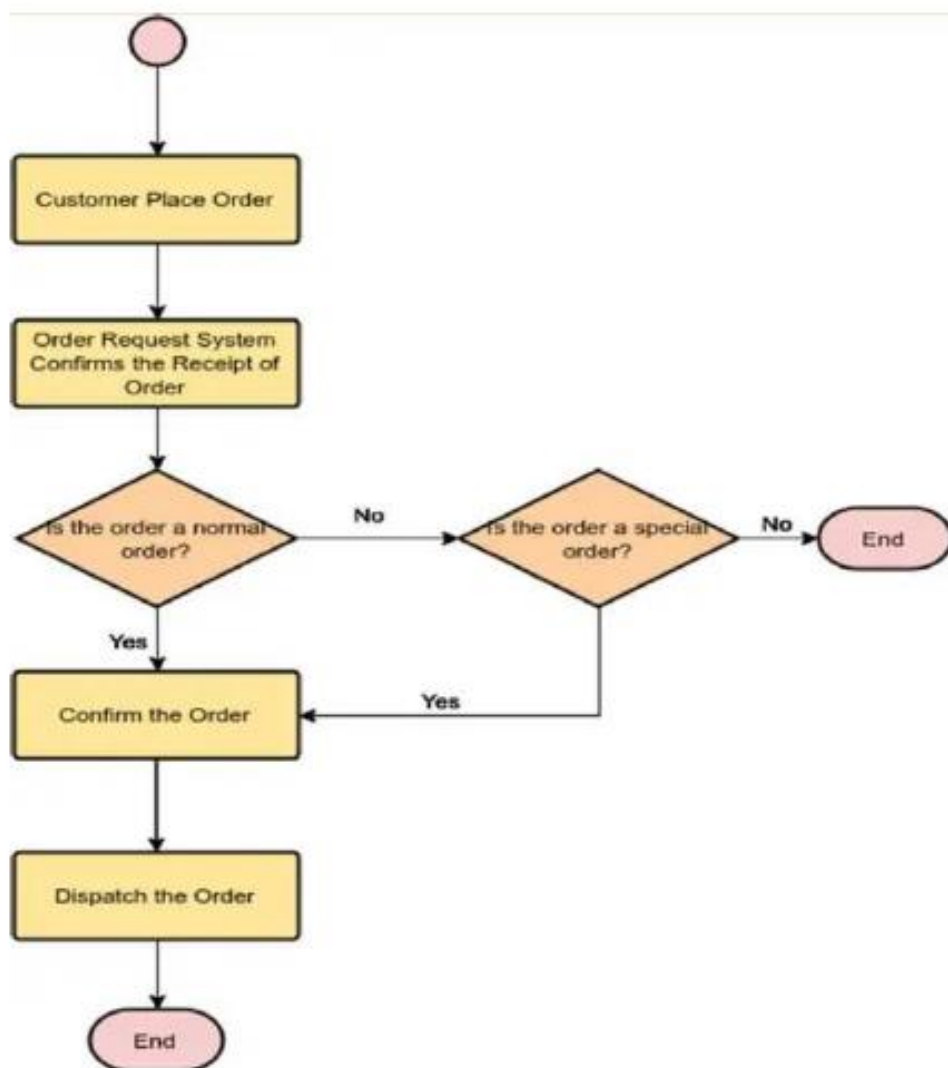
An online food ordering system offers numerous strengths that revolutionize the dining experience for users and businesses alike. Foremost among these is unparalleled convenience, as users can effortlessly browse through a vast selection of restaurants and cuisines, place orders from the comfort of their homes or workplaces, and enjoy swift delivery or pickup options. The accessibility of the system ensures that users have access to diverse culinary options, catering to varying tastes and dietary preferences. Furthermore, the system streamlines the ordering process, minimizing wait times, eliminating the need for phone calls, and facilitating quick and hassle-free transactions, particularly during peak hours. These strengths collectively enhance user satisfaction, increase operational efficiency for restaurants, and contribute to the overall growth and success of the online food ordering ecosystem.

## **FUTURE ENHANCEMENTS**

Future enhancements in online food ordering systems may include advancements in technology and user experience to further improve convenience, efficiency, and personalization. Integration of artificial intelligence (AI) and machine learning algorithms can enhance recommendation systems, providing personalized suggestions based on user preferences, order history, and contextual factors such as time of day and location. The system can implement a feature which is real time notification from the mobile phone application to the desk. This feature enables customer to request customer service through using the mobile application rather than verbally call restaurant staff to approach them. In addition, the mobile application also can implement a feature that will allow customer to update the food serve status. Additionally, implementing predictive analytics can optimize inventory management and demand forecasting for restaurants, reducing food waste and ensuring timely availability of popular dishes. Enhanced mobile applications with augmented reality (AR) features can offer immersive dining experiences, allowing users to visualize menu items in 3D and view virtual restaurant environments before placing orders. Furthermore, advancements in delivery logistics, such as drone and autonomous vehicle delivery options, can further streamline the delivery process, reduce delivery times, and enhance customer satisfaction. These future enhancements aim to continuously innovate and enhance the online food ordering experience, meeting evolving user expectations and driving growth in the industry.

## DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation of the flow of data within a system or process. In the context of an online food ordering system, a Data Flow Diagram (DFD) is a visual representation that illustrates the flow of data within the system. It depicts how data moves between various processes, entities, and data stores, providing a high-level overview of the system's functionality and data interactions.



**DFD Diagram for Online Food Ordering System**

### 1. External Entities:

- **Users:** Represented as external entities, users interact with the system by submitting orders, browsing menus, and providing feedback.
- **Restaurants:** External entities that provide menu listings, receive orders, and update order statuses.

### 2. Processes:

- **Order Placement:** This process involves users browsing menus, selecting items, customizing orders, and submitting them for processing.
- **Order Processing:** Once an order is placed, this process verifies payment, updates inventory, notifies the restaurant, and generates order confirmations for users.
- **Menu Management:** This process involves updating menu listings, adding new dishes, adjusting prices, and managing availability based on inventory levels.
- **Feedback Handling:** This process manages user feedback and reviews, collecting and storing feedback data for analysis and improvement.

### 3. Data Stores:

- **User Database:** Stores user profiles, preferences, order history, and authentication credentials.
- **Menu Database:** Stores information about restaurant menus, including dish names, descriptions, prices, and availability.
- **Order Database:** Stores order details, including items ordered, quantities, prices, delivery addresses, and order statuses.
- **Feedback Database:** Stores user feedback, ratings, reviews, and comments associated with orders and restaurant experiences.

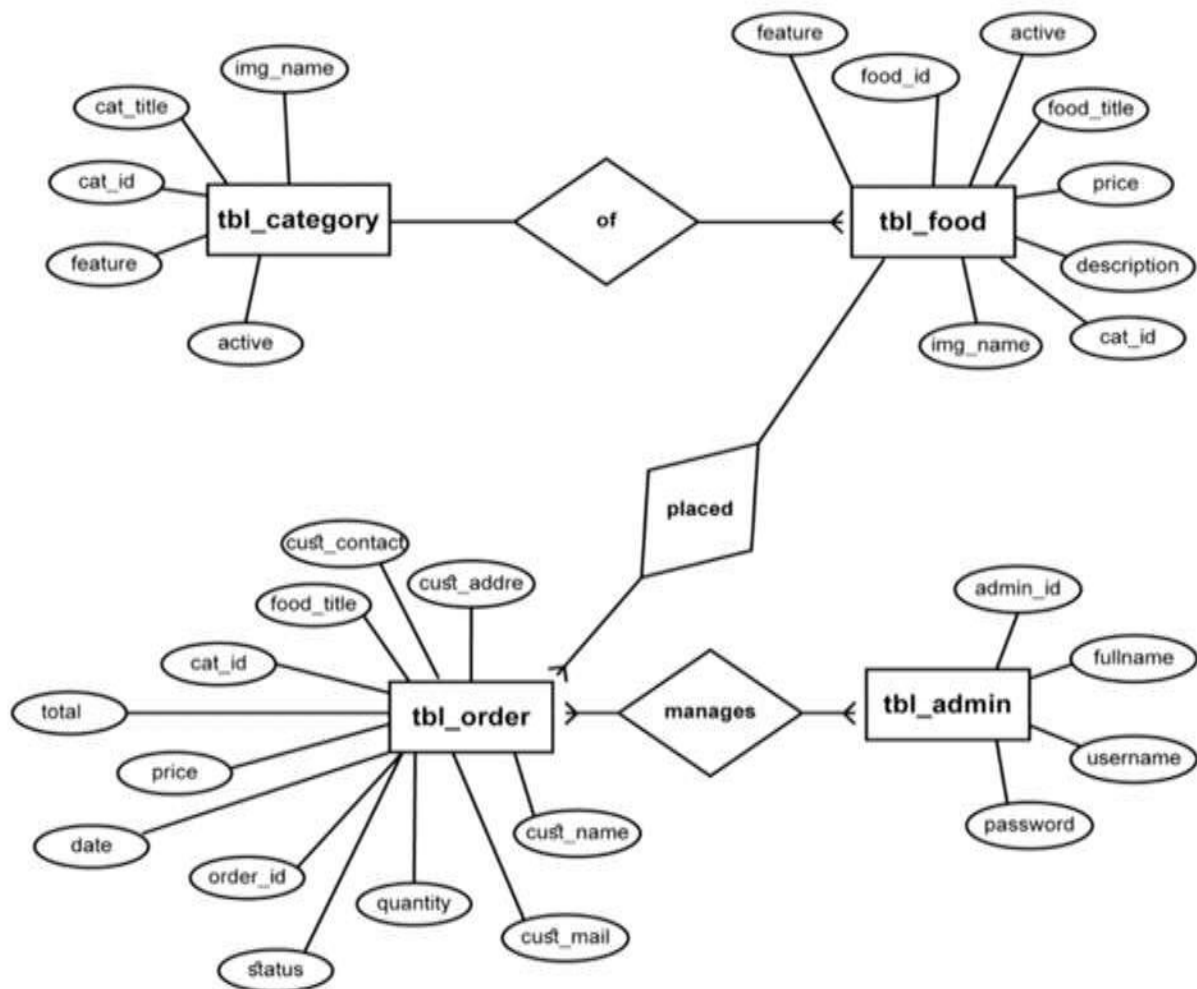


#### 4. Data Flows:

- **Order Data Flow:** Represents the flow of order information from users to the system, including order details, preferences, and delivery instructions.
- **Menu Data Flow:** Represents the flow of menu information from restaurants to the system, including menu listings, dish details, and pricing.
- **Feedback Data Flow:** Represents the flow of feedback information from users to the system, including ratings, reviews, and comments.
- **Notification Data Flow:** Represents the flow of notification information from the system to users and restaurants, including order confirmations, updates, and alerts.

## E-R DIAGRAM

In the context of an online food ordering system, an Entity-Relationship (E-R) diagram is a visual representation that depicts the entities, attributes, and relationships within the system's database. It illustrates the structure of the database and how different entities are related to each other.



**E-R Diagram for Online Food Ordering System**

## 1. Entities:

- **User:** Represents individuals who interact with the system by browsing menus, placing orders, and providing feedback. Attributes may include user ID, name, contact information, and preferences.
- **Restaurant:** Represents establishments that offer food items for ordering. Attributes may include restaurant ID, name, location, contact information, and operating hours.
- **Menu Item:** Represents individual dishes or items available for ordering from restaurants. Attributes may include item ID, name, description, price, and availability status.
- **Order:** Represents a transaction initiated by a user to purchase food items from a restaurant. Attributes may include order ID, user ID, restaurant ID, order status, total amount, and delivery details.
- **Feedback:** Represents user feedback and reviews submitted after completing an order. Attributes may include feedback ID, user ID, restaurant ID, rating, review text, and timestamp.

## 2. Relationships:

- **User-Order:** Represents the relationship between users and orders, indicating that a user can place multiple orders, but each order is associated with a single user.
- **Restaurant-Menu Item:** Represents the relationship between restaurants and menu items, indicating that a restaurant can offer multiple menu items, but each menu item is associated with a single restaurant.
- **Order-Menu Item:** Represents the relationship between orders and menu items, indicating that an order can include multiple menu items, and each menu item can be part of multiple orders.

- **User-Feedback:** Represents the relationship between users and feedback, indicating that a user can submit multiple feedback entries, but each feedback entry is associated with a single user.

### **3. Cardinality and Constraints:**

- Cardinality notations such as "one-to-many" and "many-to-many" depict the relationship between entities, indicating the number of instances of one entity that can be associated with another entity.
- Constraints such as "mandatory" or "optional" specify whether a relationship is required or optional, and whether it can exist independently of other entities.

## PURPOSE

The purpose of an online food ordering system project are multifaceted, aimed at enhancing the overall dining experience for both users and restaurant owners. The project aims to provide users with a convenient and accessible platform to browse through a diverse range of restaurants and cuisines, place orders, and arrange delivery or pickup options from the comfort of their homes or workplaces. This convenience factor is essential in today's fast-paced world, catering to busy lifestyles and offering a seamless solution for satisfying culinary cravings. Secondly, the project seeks to streamline the ordering process, minimizing wait times, eliminating the need for phone calls, and facilitating quick and hassle-free transactions, thereby enhancing operational efficiency for restaurants and reducing errors in order processing.

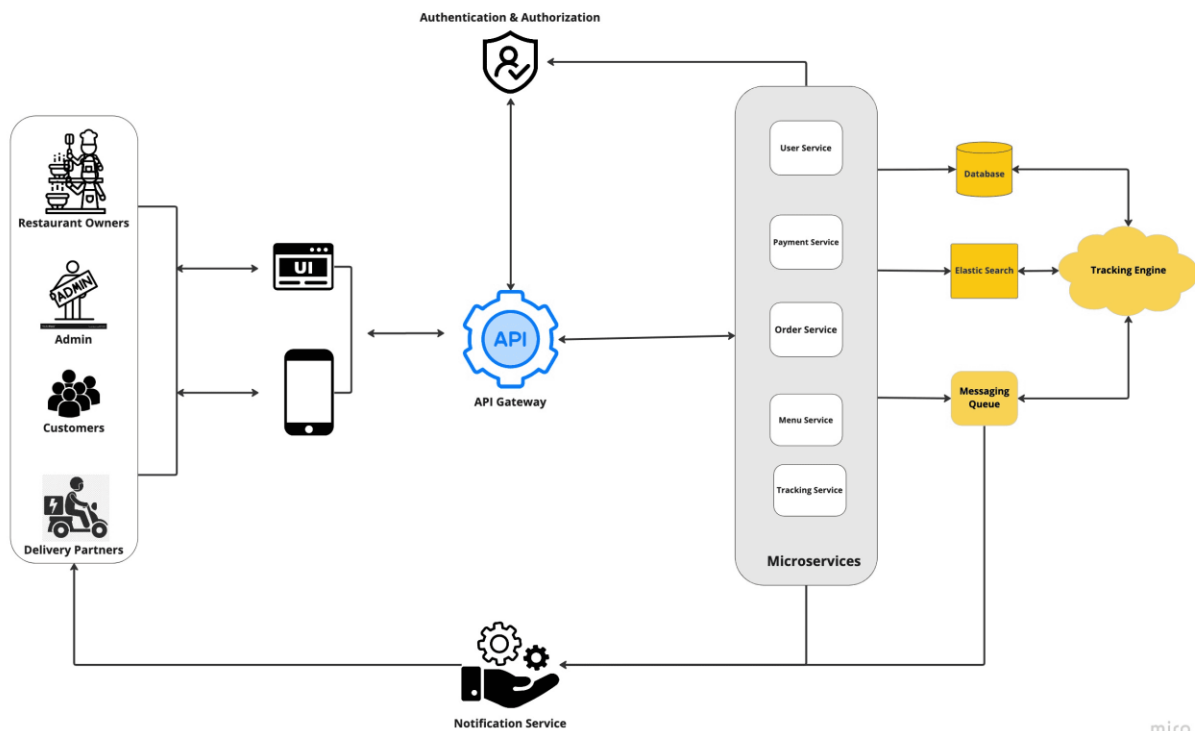
1. **Convenience:** The online food ordering system project aims to offer users a convenient way to browse menus, place orders, and arrange delivery or pickup from a wide range of restaurants and cuisines. By providing an easy-to-use platform accessible from any device with internet connectivity, the project ensures that users can satisfy their culinary cravings without the hassle of visiting restaurants or making phone calls.
2. **Efficiency:** One of the primary purposes of the project is to streamline the ordering process, reducing wait times and errors in order processing. By digitizing the ordering process, the project minimizes the manual effort required by both users and restaurant staff, leading to faster transactions, improved order accuracy, and enhanced operational efficiency for restaurants.
3. **Transparency:** The project seeks to foster transparency and trust by providing users with comprehensive information about restaurants, menus, prices, delivery options, and order statuses. By offering transparency at every stage of the ordering process, the project ensures that users can make informed decisions and trust the platform for their dining needs.
4. **Accessibility:** Another key purpose of the project is to make food ordering accessible to a wide range of users, including those with disabilities or limited mobility. By providing accessible interfaces and options for

customization, the project ensures that all users can easily navigate the platform and place orders without barriers.

- 5. Innovation:** The project aims to innovate the dining experience by leveraging technology such as artificial intelligence (AI), machine learning, and augmented reality (AR). By introducing features such as personalized recommendations, predictive analytics for inventory management, and immersive dining experiences, the project seeks to stay ahead of trends and provide users with unique and engaging experiences.
- 6. Business Growth:** Ultimately, the project aims to drive business growth for restaurants by expanding their reach, increasing order volumes, and improving customer satisfaction. By providing restaurants with a reliable platform to showcase their offerings and efficiently manage orders, the project contributes to the success and sustainability of the food service industry.

## SYSTEM ARCHITECTURE

In the context of an online food ordering system, the system architecture refers to the high-level structure and components that enable the functionality and operation of the platform. The system architecture typically comprises several layers, each serving a specific purpose and interacting with other layers to fulfill user requests and process data effectively.



**System Architecture of Online Food Ordering System**

- 1. Presentation Layer:** This layer represents the user interface of the online food ordering system, including web or mobile applications used by customers to browse restaurants, view menus, place orders, and track deliveries. It focuses on providing an intuitive and responsive interface that enhances user experience and facilitates interaction with the system.
- 2. Application Layer:** The application layer contains the business logic and functionality of the online food ordering system. It handles user requests, processes orders, manages restaurant listings and menus, handles payment transactions, and coordinates interactions between different system components. This layer implements the core features of the platform and ensures smooth operation of the ordering process.
- 3. Data Layer:** The data layer is responsible for managing and storing data related to users, restaurants, menus, orders, payments, and other relevant information. It typically includes a relational database management system (RDBMS) such as MySQL, PostgreSQL, or MongoDB, where data is organized, stored, and retrieved efficiently. The data layer ensures data integrity, security, and scalability to support the growing needs of the platform.
- 4. Integration Layer:** The integration layer facilitates communication and integration with external systems and services, such as payment gateways, delivery logistics providers, and third-party APIs. It enables seamless interactions between the online food ordering system and external entities, ensuring smooth operation and extending the platform's functionality through integrations with external services.
- 5. Infrastructure Layer:** The infrastructure layer encompasses the hardware, network infrastructure, and cloud services that support the operation of the online food ordering system. It includes servers, storage systems, networking components, and cloud platforms such as AWS, Azure, or Google Cloud, which provide the computing resources and scalability needed to host and run the platform reliably and securely.



## SNAPSHOTS OF THE PROJECT

### HOME PAGE

Main landing page where users can access key features such as browsing restaurants, exploring menus, placing orders, viewing promotions, etc.



### CUSTOMER REGISTRATION PAGE

Creating new user accounts, enabling customers to sign up by providing their personal information such as name, email address, delivery address, and contact details.

yum-yum 😊
home about menu orders contact
🔍 🛒(0) 👤

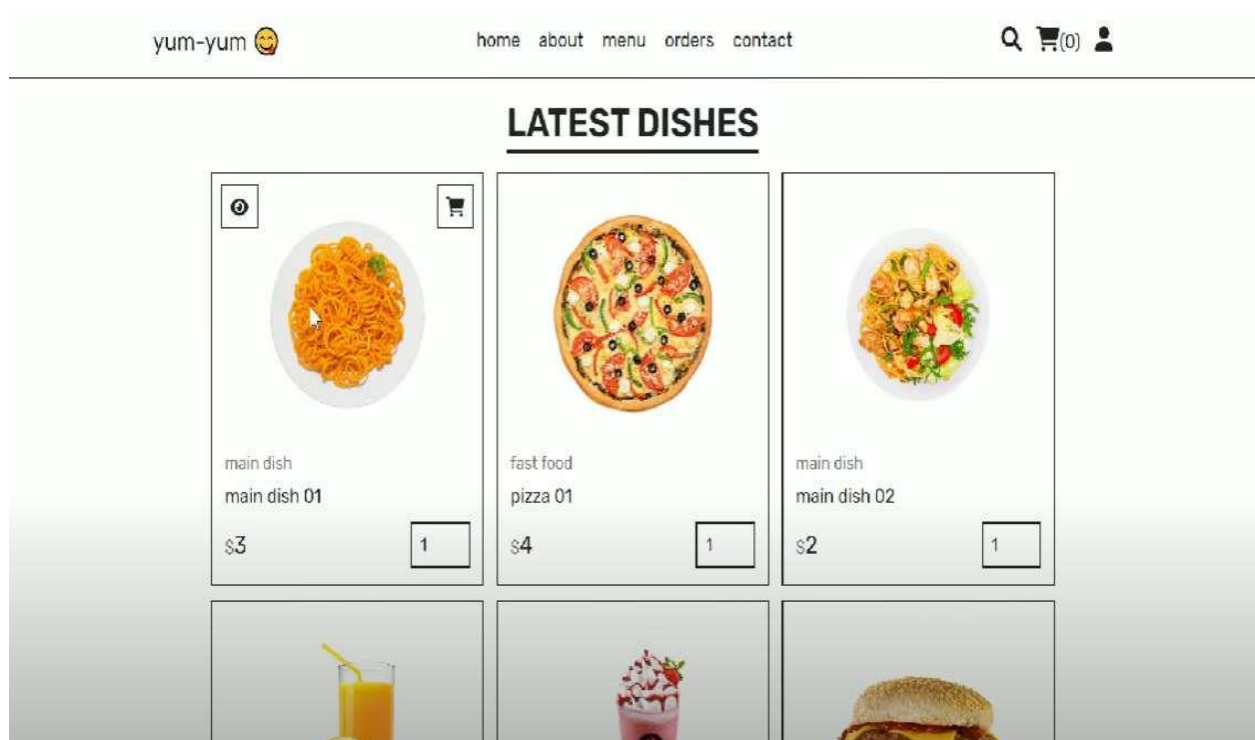
### REGISTER NOW

Register Now

already have an account? [login now](#)

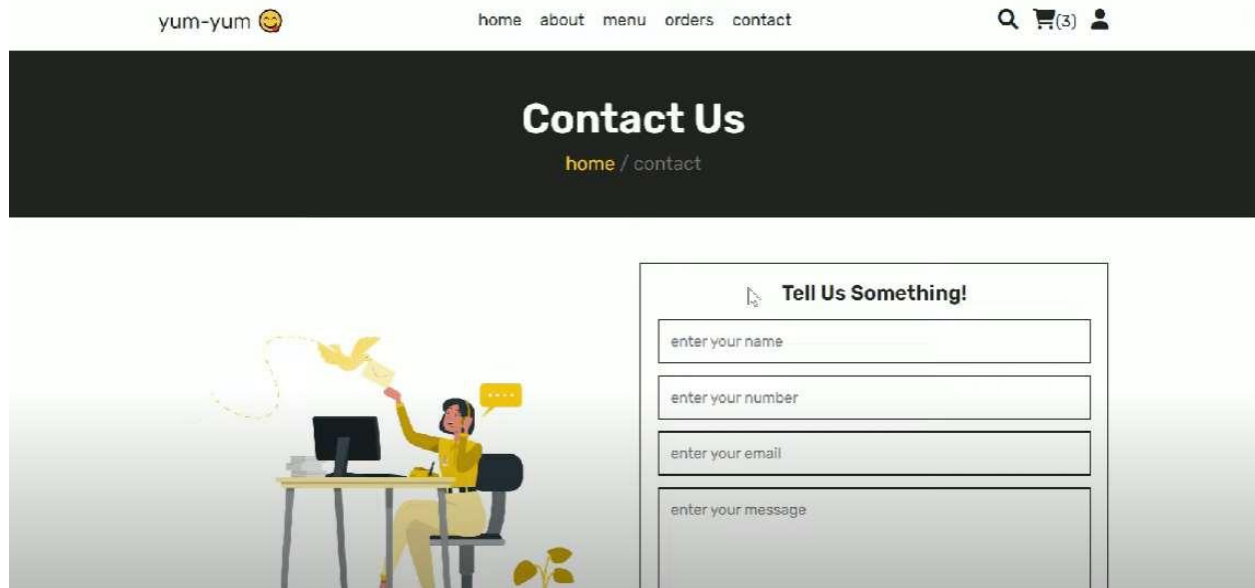
## MENU PAGE

Displaying the available food items and offerings from participating restaurants, allowing customers to browse through various categories, view detailed descriptions, images, etc.



## CONTACT US PAGE

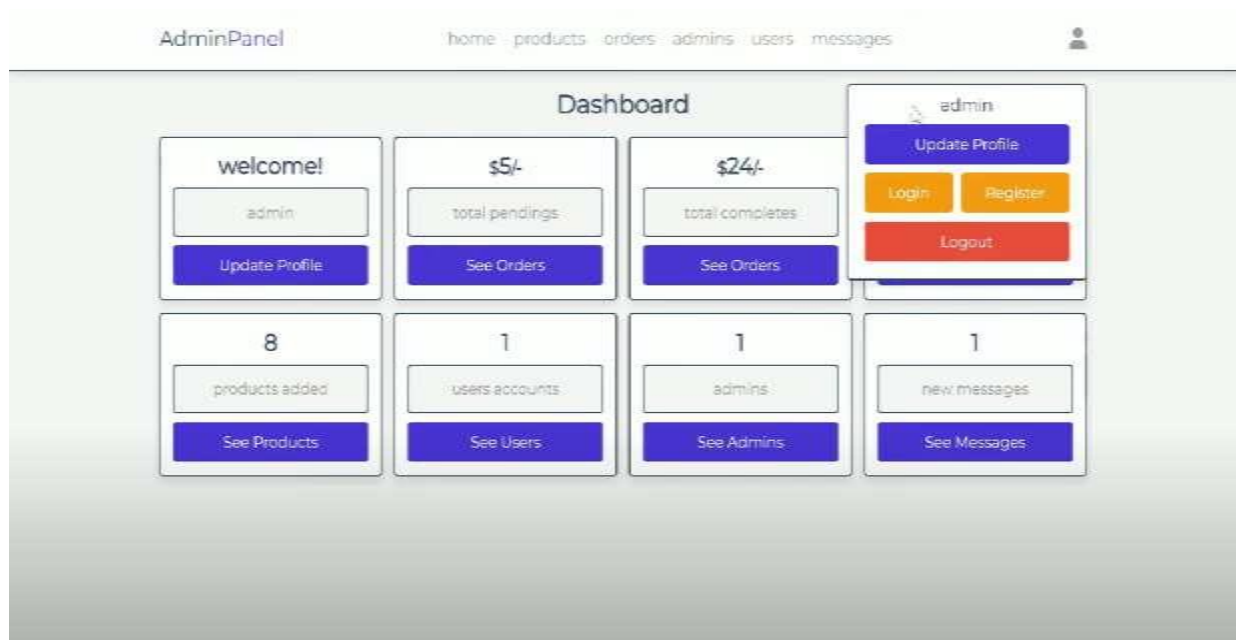
A dedicated channel for customers to reach out to the platform's support team or customer service representatives.



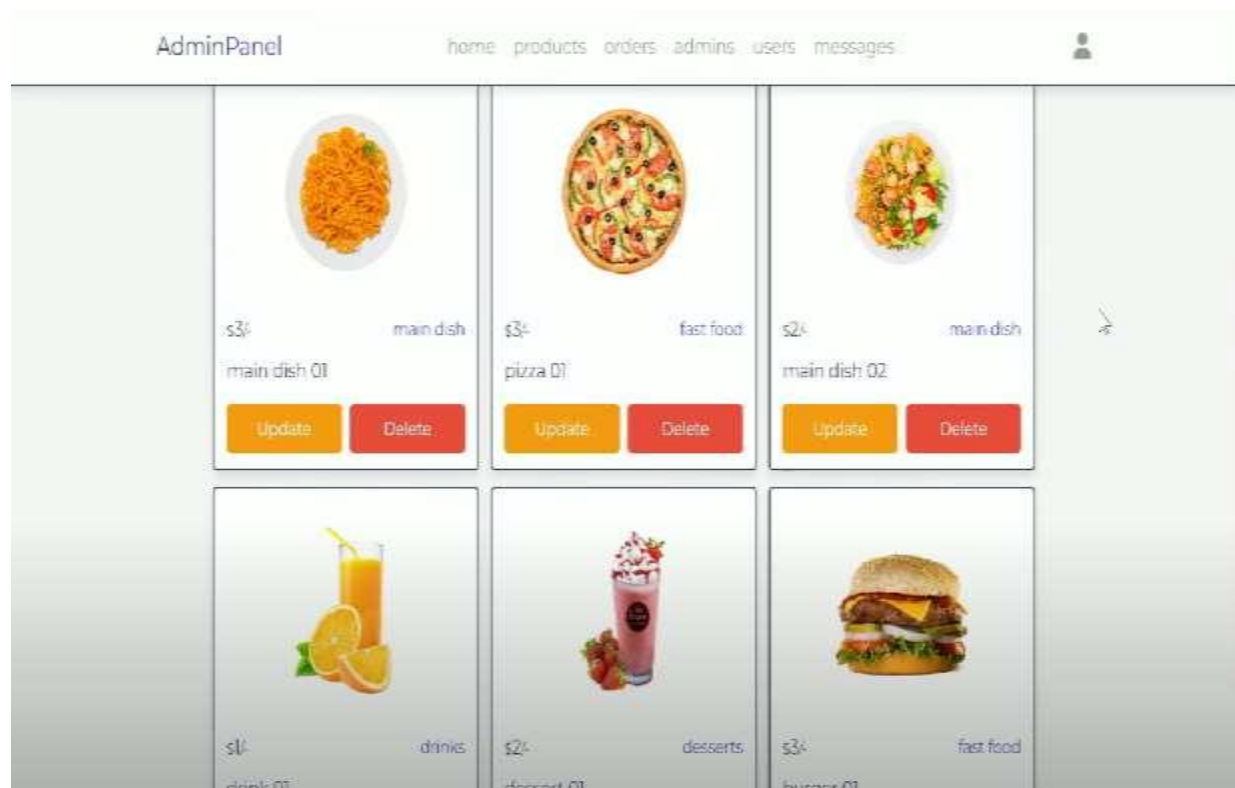
The screenshot shows the 'Contact Us' page of a website. At the top, there is a navigation bar with the logo 'yum-yum' and a smiley face icon, followed by links: 'home', 'about', 'menu', 'orders', and 'contact'. On the right side of the navigation bar, there are icons for a search magnifying glass, a shopping cart with '(3)' items, and a user profile icon. Below the navigation bar is a dark grey header with the text 'Contact Us' in white, and a breadcrumb trail 'home / contact' in yellow. The main content area features an illustration on the left of a woman in a yellow uniform sitting at a desk with a computer, holding a yellow envelope. To the right of the illustration is a contact form titled 'Tell Us Something!'. The form contains four input fields: 'enter your name', 'enter your number', 'enter your email', and 'enter your message'.

## ADMIN PANEL

Admin Panel for managing various aspects of the platform, including menu updates, order management, user accounts, etc.



The screenshot shows the 'AdminPanel' dashboard. At the top, there is a navigation bar with the text 'AdminPanel' and a series of links: 'home', 'products', 'orders', 'admins', 'users', and 'messages'. On the right side of the navigation bar, there is a user profile icon. Below the navigation bar is a dashboard titled 'Dashboard'. The dashboard contains several widgets. On the left, there is a 'welcome!' widget for the user 'admin' with an 'Update Profile' button. Next to it is a widget showing '\$5/-' for 'total pendings' with a 'See Orders' button. To the right of that is a widget showing '\$24/-' for 'total completes' with a 'See Orders' button. On the far right, there is a sidebar for the user 'admin' with buttons for 'Update Profile', 'Login', 'Register', and 'Logout'. Below the 'welcome!' widget is a widget showing '8' for 'products added' with a 'See Products' button. Next to it is a widget showing '1' for 'users accounts' with a 'See Users' button. To the right of that is a widget showing '1' for 'admins' with a 'See Admins' button. On the far right of the dashboard is a widget showing '1' for 'new messages' with a 'See Messages' button.



## **CONCLUSION**

After a decade, the advancement and innovation of technology helps people to manage their task easily and efficiently from just a click of a button. In many other industry area have been used management system to assist their business grow long time ago, therefore it is a trend that cause F&B industry to make use of a management system for their business. At the end of this project the system can reduce and replace the human manpower task, reduce the time consume for each transaction and generate report for further management purpose by fully utilizing the system. Development of an online food ordering system represents a significant step forward in modernizing the dining experience for users and restaurant owners alike. Through the convenience of digital platforms, users can effortlessly explore a vast array of culinary options, place orders, and arrange delivery or pickup with ease. This system streamlines the ordering process, minimizing wait times, eliminating errors, and fostering transparency in transactions. Moreover, it provides restaurants with a platform to expand their reach, increase order volumes, and enhance customer satisfaction. The project's success hinges on robust infrastructure, intuitive user interfaces, seamless integration with external services, and a commitment to continuous improvement. As technology evolves and user preferences shift, the online food ordering system must adapt and innovate to meet evolving demands and remain competitive in the dynamic food service industry. Overall, the project signifies a transformative shift towards digitization and convenience in the realm of dining, promising a more efficient, accessible, and satisfying experience for all stakeholders involved.

## **FUTURE WORK**

Future work for the online food ordering system involves continual innovation and adaptation to meet evolving user preferences and technological advancements. This includes enhancing personalization through advanced recommendation algorithms, integrating emerging technologies such as AI and AR to enhance user experience, expanding delivery options for greater convenience, ensuring accessibility for users with disabilities, optimizing sustainability practices, expanding into new markets for global growth, continuously optimizing platform performance, and leveraging data analytics for insights-driven decision-making. By focusing on these areas, the online food ordering system can remain competitive, responsive, and relevant in the ever-changing landscape of the food service industry.

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